Structural j-formula and upper bound for $\lambda = (4, 3, 1^q)$, $q \geq 3$: the first $r = 2$ same-row family beyond $(4, 2, 1^*)$

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Abstract

The May-13 multi-corner paper (`2026-05-13-multi-corner-dimension.tex`) conjectured a closed-form dimension $\dim B^{(\lambda)}_{\ell+1} V_\lambda = f^{\lambda^{(\geq 2)}}$ in the stable regime. The two-corner family $\lambda = (4, 3, 1^q)$ has decoration shape $\lambda^{(\geq 2)} = (3, 2)$, predicting $\dim = f^{(3,2)} = 5$ (verified computationally at $q \in \{3, 4\}$).

We prove the matching statement at the j-formula level: for every $q \geq 3$,

$$B^{(\lambda)}_{j_0} V_\lambda \;\subseteq\; E^-_1|_{V_\lambda},$$

and the upper bound

$$\dim B^{(\lambda)}_{j_0} V_\lambda \;\leq\; 5.$$

Both statements are *fully structural* at $q \geq 5$. At $q \in \{3, 4\}$, the four-branch reduction itself is structural (the two zero pieces μ_C-leakage and same-row-dies are structural at $q \geq 3$), but the two non-zero Φ-summand inputs come from families $(3, 3, 1^{q-1})$ and $(4, 2, 1^{q-1})$ at sub-shape sizes 8 or 9, where the relevant j-formulas and dim formulas have only computational verification (the structural ranges of SRD-331 and SRD-421 cut in at sub-shape size ≥ 10). Combined with the computational lower bound $\dim B \geq 5$, this gives $\dim B = 5$ on every $q \in \{3, 4, 5, \ldots\}$ in the ranges tested.

Proof strategy. Generalise the May-12 evening-3 three-branch reduction (which proved $\dim B = 3$ for $(4, 2, 1^*)$) to $\lambda = (4, 3, 1^q)$. The shape has $r = 2$ length-preserving corners and one same-row pair at row 2; the Phase A decomposition has *four* summands rather than three:

$$E^+_{n-1}|_{V_\lambda} = \;\Phi_A(V_{\mu_A}) \;\oplus\; \Phi_B(V_{\mu_B}) \;\oplus\; \Phi_C(V_{\mu_C}) \;\oplus\; S,$$

with $\mu_A = (3, 3, 1^{q-1})$, $\mu_B = (4, 2, 1^{q-1})$, $\mu_C = (3, 2, 1^q)$ (each on $n-2$ letters), and S the same-row part at row 2 with sub-shape $\nu_2 = (4, 1^{q+1})$. Each summand contributes precisely:

- $\mu_A = (3, 3, 1^{q-1})$: $\dim B^{(\mu_A)} = 2$ (May-13 same-row-dies-$(3, 3, 1^*)$, structural at $q - 1 \geq 4$).
- $\mu_B = (4, 2, 1^{q-1})$: $\dim B^{(\mu_B)} = 3$ (May-13 same-row-dies-$(4, 2, 1^*)$ + May-12-Eve3, structural at $|\mu_B| \geq 10$, i.e. $q - 1 \geq 4$).
- $\mu_C = (3, 2, 1^q)$: 0 by j-formula leakage (May-12 today j-formula plus the trailing $(T_1{+}1)$ argument).
- S: 0 by the general same-row-dies framework (Theorem 14 of May-13 same-row-dies-$(4, 2, 1^*)$) applied with the hook j-formula on $\nu_2 = (4, 1^{q+1})$ (May-13 Shift Lemma all-hooks).

Upper bound: $\dim \leq 2 + 3 + 0 + 0 = 5$. The E^-_1 containment propagates through every piece because the outer factors $(T_j{+}1)$, $j \geq j_0 - 1 = q + 2 \geq 5$, commute with T_1.

Significance. This is the first $r = 2$ same-row family beyond $\lambda = (4, 2, 1^*)$. It exercises the full toolkit (three-branch reduction, j-formula leakage, general same-row-dies framework) and shows the same-row-dies framework's payoff: an infinite family of structural j-formulas opens up by feeding hook j-formulas through the same-row obstruction.

The matching structural *lower bound* (constructing five linearly independent vectors in B via position-of-n / position-of-$(n-1)$ separators) is deferred to a follow-up paper; here we record only the upper bound and the E_1^- containment.

1 Setup

Conventions follow the May-12 evening-3 paper (`2026-05-12-evening-4-2-1n-dim3.tex`, henceforth **May-12-Eve3**), the May-13 same-row-dies-$(4, 2, 1^*)$ paper (`2026-05-13-same-row-dies-421.tex`, henceforth **SRD-421**), and the May-13 Shift Lemma paper (`2026-05-13-shift-lemma-all-hooks.tex`, henceforth **Shift**). $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \ldots, T_{n-1}$ obeying $(T_i - q)(T_i + 1) = 0$ and braid. V_λ is the seminormal cell module for $\lambda \vdash n$, with basis $\{v_T : T \in \mathrm{SYT}(\lambda)\}$ and asymmetric Murphy normalisation ($b' = 1$). $E_j^\pm \subseteq V_\lambda$ are the q- and (-1)-eigenspaces of T_j. For $1 \le k \le n$,

$$R'_k \;:=\; (T_{k-1}{+}1)(T_{k-2}{+}1)\cdots(T_1{+}1) \;\in\; H_q(S_n),$$

with $R'_1 = 1$. For $1 \le a \le b \le n$, $P_{a,b} := R'_a R'_{a+1} \cdots R'_b$ (empty product $= 1$ if $a > b$). The j-image is

$$B_j^{(\lambda)} V_\lambda \;:=\; P_{j,n} V_\lambda \;=\; R'_j R'_{j+1} \cdots R'_n V_\lambda, \qquad j_0(\lambda) \;:=\; \ell(\lambda) + 1.$$

Throughout this paper. $\lambda := (4, 3, 1^q)$ with $q \ge 3$, $n := q + 7$, $\ell := \ell(\lambda) = q + 2$, $j_0 := j_0(\lambda) = q + 3 = n - 4$.

Removable corners of λ.

$c_1 := (1, 4)$ (length-preserving), $\quad c_2 := (2, 3)$ (length-preserving), $\quad c_3 := c_*(\lambda) = (q{+}2, 1)$ (column-1 bottom).

Length-preservation of c_1: $\lambda_2 = 3 < 4 = \lambda_1$. Length-preservation of c_2: $\lambda_3 = 1 < 3 = \lambda_2$. $\lambda_\ell = 1$, so c_3 is the removable column-1 bottom.

Same-row pairs of λ. Row 1: needs $\lambda_2 \le \lambda_1 - 2 = 2$, but $\lambda_2 = 3$; *no same-row pair at row* 1. Row 2: needs $\lambda_3 \le \lambda_2 - 2 = 1$, and $\lambda_3 = 1$; *same-row pair at row* 2, cells $\{(2, 2), (2, 3)\}$. Rows ≥ 3 have length 1; no same-row pairs.

Inner partitions on $n{-}2$ letters. For each corner pair $X = \{c_i, c_j\}$ with $i < j$, set $\mu_X := \lambda \setminus X$:

$$\begin{aligned}
\mu_A &:= \lambda \setminus \{c_1, c_3\} = (3, 3, 1^{q-1}), && \text{(top corner } c_1 \text{ paired with bottom } c_3)\\
\mu_B &:= \lambda \setminus \{c_2, c_3\} = (4, 2, 1^{q-1}), && \text{(middle corner } c_2 \text{ paired with bottom } c_3)\\
\mu_C &:= \lambda \setminus \{c_1, c_2\} = (3, 2, 1^q), && \text{(both length-pres corners paired).}
\end{aligned}$$

All three have $|\mu_X| = q + 5 = n - 2$, and

$$\ell(\mu_A) = \ell(\mu_B) = q + 1, \qquad \ell(\mu_C) = q + 2 = \ell(\lambda).$$

Sharp indices: $j_0(\mu_A) = j_0(\mu_B) = q + 2 = j_0(\lambda) - 1$ and $j_0(\mu_C) = q + 3 = j_0(\lambda)$.

Same-row sub-shape. $\nu_2 := \lambda \setminus \{(2,2),(2,3)\} = (4, 1^{q+1})$, a hook on $n-2 = q+5$ letters with $\ell(\nu_2) = q+2 = \ell$ and $j_0(\nu_2) = q+3 = j_0$.

2 Known ingredients

Theorem 1 (Phase A endpoint, May-19)**.** *For any $H_q(S_n)$-module V, $R'_n V = E^+_{n-1}(V)$.*

Theorem 2 (Corner-pair embedding Φ_X, May-20 + SRD-421)**.** *For each ordered corner pair $X = (c, c')$ of λ with $|d_X| \geq 2$ (d_X the axial distance), there is a $\mathbb{Q}(q)$-linear injection*

$$\Phi_X : V_{\mu_X} \hookrightarrow V_\lambda$$

with the following properties:

1. Construction. *For each SYT T' of μ_X, $\Phi_X(v_{T'})$ is the T_{n-1} $+q$-eigenvector of the 2-block $\{v_{T^X_A}, v_{T^X_B}\}$ in V_λ where T^X_A, T^X_B extend T' by placing the letters $\{n-1, n\}$ at the cells of X (in opposite orders).*
2. Equivariance. *$T_i \Phi_X(v) = \Phi_X(T_i v)$ for every $v \in V_{\mu_X}$ and every $1 \leq i \leq n-3$.*
3. Image. *$\Phi_X(V_{\mu_X})$ is the span of T_{n-1} $+q$-eigenvectors in the corner-pair 2-blocks for the pair X.*
4. E^-_1-preservation. *For $n \geq 4$, Φ_X identifies $E^-_1|_{V_{\mu_X}}$ with $\Phi_X(V_{\mu_X}) \cap E^-_1|_{V_\lambda}$.*

Lemma 3 (Phase A decomposition for λ)**.** *For $\lambda = (4, 3, 1^q)$ at $q \geq 1$,*

$$E^+_{n-1}|_{V_\lambda} = \Phi_A(V_{\mu_A}) \oplus \Phi_B(V_{\mu_B}) \oplus \Phi_C(V_{\mu_C}) \oplus S, \tag{1}$$

where S is the span of v_T for SYTs T with $T(2,2) = n-1$, $T(2,3) = n$.

Proof. By Theorem 1, $E^+_{n-1}|_{V_\lambda} = R'_n V_\lambda$, and the $+q$-eigenvectors of T_{n-1} in the seminormal basis of V_λ arise from two configurations of the letters $\{n-1, n\}$ in an SYT (see SRD-421, Lemma 3/Lemma 12 there):

(i) $\{n-1, n\}$ at a corner pair $\{c, c'\}$ in different rows and columns, giving a non-degenerate T_{n-1}-2-block whose $+q$-eigenvector is $\Phi_{\{c,c'\}}(v_{T'})$ for the restriction $T' = T|_{\mu_{\{c,c'\}}}$. The three corner pairs are $\{c_1, c_3\} = A$, $\{c_2, c_3\} = B$, $\{c_1, c_2\} = C$.

(ii) $\{n-1, n\}$ in a same-row pair, giving v_T itself as a $+q$-eigenvector (Hoefsmit rule: $T_{n-1} v_T = q v_T$ when $n-1, n$ are in the same row). The unique same-row pair of λ is $\{(2,2),(2,3)\}$ at row 2.

The four families have disjoint seminormal supports (each is characterised by where $\{n-1, n\}$ sit), so the sum is direct. The total dimension is $\sum_X f^{\mu_X} + f^{\nu_2}$, which equals the total number of $+q$-eigenvectors of T_{n-1} in V_λ. □ □

Theorem 4 (μ_B: dim formula and j-formula on $(4, 2, 1^*)$, May-12-Eve3 + SRD-421)**.** *For $\mu_B = (4, 2, 1^{q-1})$:*

1. $\dim B^{(\mu_B)}_{j_0(\mu_B)} V_{\mu_B} = 3$ structurally *at $|\mu_B| \geq 10$ (i.e. $q-1 \geq 4$, equivalently $q \geq 5$); for $q - 1 \in \{2, 3\}$, $\dim = 3$ is established computationally (mod $P = 100\,003$, $Q = 11$).*
2. $B^{(\mu_B)}_{j_0(\mu_B)} V_{\mu_B} \subseteq E^-_1|_{V_{\mu_B}}$ structurally *at $|\mu_B| \geq 10$ (i.e. $q \geq 5$); for smaller $|\mu_B|$ it is established computationally.*

Remark 5. The structural range $|\mu_B| \geq 10$ inside SRD-421 comes from sub-shape dependencies: the upper bound and E_1^- containment on $(4,2,1^*)$ rely on the j-formula on $(3,2,1^*)$ (May-12 today, structural at size ≥ 8). Same-row dies for $(4,2,1^*)$ alone is structural at all $q-1 \geq 2$, but does not by itself give the E_1^- containment on μ_B.

Theorem 6 (μ_A: same-row dies for $(3,3,1^*)$ + dim formula)**.** *For $\mu_A = (3,3,1^{q-1})$:*

1. (SRD-331, May-13). *The j-formula $B^{(\mu_A)}_{j_0(\mu_A)} V_{\mu_A} \subseteq E_1^- |_{V_{\mu_A}}$ holds for every $q-1 \geq 2$ (i.e. $q \geq 3$).*
2. (SRD-331, May-13.) $\dim B^{(\mu_A)}_{j_0(\mu_A)} V_{\mu_A} = 2$ *structurally for $q-1 \geq 4$ (i.e. $q \geq 5$); for $q-1 \in \{2,3\}$ the $\dim = 2$ is established computationally (mod $P = 100\,003$, $Q = 11$).*

Theorem 7 (μ_C at sharp index: j-formula on $(3,2,1^*)$, May-12-today)**.** *For $\mu_C = (3,2,1^q)$ on $n-2 = q+5$ letters with $q \geq 1$ (so $n-2 \geq 6$),*

$$B^{(\mu_C)}_{j_0(\mu_C)} V_{\mu_C} \;\subseteq\; E_1^- |_{V_{\mu_C}} \;.$$

Theorem 8 (General same-row-dies framework, SRD-421 Theorem 14)**.** *Let $\widetilde{\lambda} \vdash N$ admit a same-row pair at row i, set $\widetilde{\nu}_i := \widetilde{\lambda} \setminus \{(i, \widetilde{\lambda}_i - 1), (i, \widetilde{\lambda}_i)\}$, $\widetilde{\ell} := \ell(\widetilde{\lambda})$, $\widetilde{j}_0 := \widetilde{\ell} + 1$, and $\widetilde{S}_i$ the corresponding same-row span. Assume $\widetilde{\lambda}_i \geq 3$ (so $\ell(\widetilde{\nu}_i) = \widetilde{\ell}$) and $B^{(\widetilde{\nu}_i)}_{j_0(\widetilde{\nu}_i)} V_{\widetilde{\nu}_i} \subseteq E_1^- |_{V_{\widetilde{\nu}_i}}$ (j-formula on $\widetilde{\nu}_i$). Then*

$$R'_{\widetilde{j}_0} \, R'_{\widetilde{j}_0+1} \cdots R'_{N-1} \, \widetilde{S}_i \;=\; 0.$$

Theorem 9 (Hook j-formula at all k, Shift Lemma)**.** *For every hook $\eta = (k, 1^m)$ on $k+m \geq 3$ letters with $k \geq 2$,*

$$B^{(\eta)}_{j_0(\eta)} V_\eta \;\subseteq\; E_1^- |_{V_\eta} \;.$$

3 The four-branch reduction

We generalise the May-12-Eve3 three-branch reduction to handle four summands and one extra commutation step.

3.1 Commutation through Φ_X

Lemma 10 (Commutation lemma)**.** *For each $X \in \{A, B, C\}$ and every $1 \leq k \leq n-1$,*

$$R'_k \, \Phi_X(v) \;=\; (T_{k-1}{+}1) \, \Phi_X\big(R'^{(\mu_X)}_{k-1} \, v\big) \qquad \text{for all } v \in V_{\mu_X}.$$

Here $R'^{(\mu_X)}_{k-1}$ is the R'_{k-1} operator in $H_q(S_{|\mu_X|}) = H_q(S_{n-2})$, viewed via the natural index inclusion. Iterating,

$$P_{j_0,n-1} \, \Phi_X(V_{\mu_X}) \;=\; (T_{j_0-1}{+}1)(T_{j_0}{+}1) \cdots (T_{n-2}{+}1) \, \Phi_X\big(P^{(\mu_X)}_{j_0-1,n-2} \, V_{\mu_X}\big). \tag{2}$$

Proof. The factor $(T_{k-1}+1)$ at the leftmost end of R'_k has the largest index $k-1 \leq n-2$. By Theorem 2(2), all factors $(T_j{+}1)$ with $j \leq n-3$ commute through Φ_X equivariantly, so the rightmost-to-second-leftmost factors of R'_k commute through Φ_X as the operator $R'^{(\mu_X)}_{k-1} = (T_{k-2}{+}1)(T_{k-3}{+}1) \cdots (T_1{+}1)$ acting inside V_{μ_X}. The leftmost factor $(T_{k-1}{+}1)$ remains outside as $(T_{k-1}{+}1)$.

Iterating: for $j_0 \le k \le n-1$,

$$P_{j_0,n-1}\,\Phi_X = R'_{j_0} R'_{j_0+1} \cdots R'_{n-1}\,\Phi_X = (T_{j_0-1}{+}1)\,\Phi_X\, R'^{(\mu_X)}_{j_0-1}\, R'_{j_0+1} \cdots R'_{n-1}.$$

The next R'_{j_0+1} commutes through Φ_X (peeling off $(T_{j_0}{+}1)$ as a left outer factor) past $R'^{(\mu_X)}_{j_0-1}$ *provided* $(T_{j_0}{+}1)$ commutes with $R'^{(\mu_X)}_{j_0-1}$ on the μ_X-side; this follows from the index gap $j_0 - (j_0-2) = 2$ between T_{j_0} and the largest index $j_0 - 2$ appearing inside $R'^{(\mu_X)}_{j_0-1}$. (More precisely: $R'^{(\mu_X)}_{j_0-1} = (T_{j_0-2}{+}1)(T_{j_0-3}{+}1)\cdots(T_1{+}1)$ has largest index j_0-2, and T_{j_0} commutes with T_j for $|j_0 - j| \ge 2$, i.e., $j \le j_0 - 2$.) Continuing in this manner, each outer factor $(T_{k-1}{+}1)$ stays outside, and the inner $R'^{(\mu_X)}_{k-1}$ accumulates into $P^{(\mu_X)}_{j_0-1,n-2}$. Hence (2). □

3.2 The four-branch reduction

Theorem 11 (Four-branch reduction)**.** *For $\lambda = (4,3,1^q)$ at every $q \ge 1$,*

$$B^{(\lambda)}_{j_0} V_\lambda \;=\; (T_{j_0-1}{+}1)(T_{j_0}{+}1)\cdots(T_{n-2}{+}1)\Big[\sum_{X\in\{A,B,C\}} \Phi_X\big(P^{(\mu_X)}_{j_0-1,n-2} V_{\mu_X}\big)\Big] \;+\; P_{j_0,n-1}\, S. \qquad (3)$$

Proof. By definition, $B^{(\lambda)}_{j_0} V_\lambda = P_{j_0,n} V_\lambda = P_{j_0,n-1}\, R'_n V_\lambda$, and by Theorem 1 together with the Phase A decomposition (Lemma 3),

$$R'_n V_\lambda \;=\; E^+_{n-1}|_{V_\lambda} \;=\; \Phi_A(V_{\mu_A}) \oplus \Phi_B(V_{\mu_B}) \oplus \Phi_C(V_{\mu_C}) \oplus S.$$

Applying $P_{j_0,n-1}$ and using linearity together with Lemma 10 on each Φ_X-summand gives (3). □

□

4 Each piece evaluated

4.1 μ_A piece: $\dim B^{(\mu_A)} = 2$ via SRD-331

By Theorem 6,

$$P^{(\mu_A)}_{j_0-1,n-2} V_{\mu_A} \;=\; B^{(\mu_A)}_{j_0(\mu_A)} V_{\mu_A} \;\subseteq\; E^-_1|_{V_{\mu_A}},$$

since $j_0(\mu_A) = q+2 = j_0 - 1$ and $n - 2 = q + 5 = |\mu_A|$. The dimension is 2 (structural at $q \ge 5$, computational at $q \in \{3,4\}$).

4.2 μ_B piece: $\dim B^{(\mu_B)} = 3$ via SRD-421

By Theorem 4,

$$P^{(\mu_B)}_{j_0-1,n-2} V_{\mu_B} \;=\; B^{(\mu_B)}_{j_0(\mu_B)} V_{\mu_B} \;\subseteq\; E^-_1|_{V_{\mu_B}},$$

since $j_0(\mu_B) = q+2 = j_0 - 1$ and $n - 2 = q + 5 = |\mu_B|$. The dimension is 3 (structural for all $q \ge 3$).

4.3 μ_C piece is zero by j-formula leakage

Lemma 12 (μ_C piece vanishes)**.** *For $q \ge 1$, $P^{(\mu_C)}_{j_0-1,n-2}\, V_{\mu_C} = 0$.*

Proof. Since $\ell(\mu_C) = q + 2 = \ell(\lambda)$, the sharp index for μ_C is $j_0(\mu_C) = q + 3 = j_0(\lambda)$. Our inner iteration ranges $j_0 - 1$ to $n - 2 = q + 5$, one step beyond the sharp index of μ_C:

$$P^{(\mu_C)}_{j_0-1,n-2} V_{\mu_C} \;=\; R'^{(\mu_C)}_{j_0-1}\, P^{(\mu_C)}_{j_0,n-2}\, V_{\mu_C} \;=\; R'^{(\mu_C)}_{q+2}\, \big(B^{(\mu_C)}_{j_0(\mu_C)} V_{\mu_C}\big).$$

By Theorem 7, $B^{(\mu_C)}_{j_0(\mu_C)} V_{\mu_C} \subseteq E_1^-|_{V_{\mu_C}}$. The operator $R'^{(\mu_C)}_{q+2} = (T_{q+1}{+}1)(T_q{+}1)\cdots(T_2{+}1)(T_1{+}1)$ has rightmost factor $(T_1{+}1)$, which is applied first to any input vector. On E_1^-, T_1 acts as -1, so $(T_1{+}1)|_{E_1^-} = 0$. Hence $R'^{(\mu_C)}_{q+2}\, E_1^-|_{V_{\mu_C}} = 0$, giving the claim. □ □

4.4 S piece vanishes by the same-row-dies framework

Lemma 13 (S piece vanishes)**.** *For $q \geq 1$, $P_{j_0,n-1}\, S = 0$.*

Proof. We apply Theorem 8 (general same-row-dies framework) with $\widetilde{\lambda} = \lambda$ and $i = 2$:

- Same-row pair at row 2: $\lambda_2 = 3 \geq 2$ and $\lambda_3 = 1 \leq \lambda_2 - 2 = 1$. ✓
- $\lambda_2 = 3 \geq 3$, so $\ell(\nu_2) = \ell(\lambda)$ (since removing two cells $(2,2),(2,3)$ leaves $(2,1)$ as the row-2 cell, preserving row 2). ✓
- $\nu_2 = (4, 1^{q+1})$ is a hook on $n - 2 = q + 5 \geq 8$ letters. By Theorem 9 (Shift Lemma all-hooks) with $k = 4$ and $m = q + 1$, $B^{(\nu_2)}_{j_0(\nu_2)} V_{\nu_2} \subseteq E_1^-|_{V_{\nu_2}}$. ✓

Hence $P_{j_0,n-1}\, S = R'_{j_0} R'_{j_0+1} \cdots R'_{n-1}\, S = 0$. □ □

5 Main theorem: j-formula and upper bound

Theorem 14 (j-formula at sharp index)**.** *For $\lambda = (4, 3, 1^q)$ at every $q \geq 3$,*

$$B^{(\lambda)}_{j_0} V_\lambda \;\subseteq\; E_1^-|_{V_\lambda}\,.$$

This is structural at $q \geq 5$; at $q \in \{3, 4\}$ it relies on the computational E_1^- containment for $\mu_A = (3, 3, 1^{q-1})$ (automatic, see Theorem 6(1) for $q \geq 3$) and for $\mu_B = (4, 2, 1^{q-1})$ at $|\mu_B| \in \{8, 9\}$ (computational input from Theorem 4(2)).

Proof. By the four-branch reduction (Theorem 11) and Lemmas 12 and 13,

$$B^{(\lambda)}_{j_0} V_\lambda \;=\; (T_{j_0-1}{+}1)(T_{j_0}{+}1)\cdots(T_{n-2}{+}1)\Big[\Phi_A\big(B^{(\mu_A)}_{j_0(\mu_A)} V_{\mu_A}\big) + \Phi_B\big(B^{(\mu_B)}_{j_0(\mu_B)} V_{\mu_B}\big)\Big].$$

Note: Lemma 12 (the μ_C-leakage) and Lemma 13 (same-row dies) are fully structural at every $q \geq 3$; the conditional pieces are the two Φ_X-summands above.

For the E_1^- containment: by Theorems 6(1) (structural at $q \geq 3$) and 4(2) (structural at $q \geq 5$; computational at $q \in \{3, 4\}$), both inner images lie in E_1^- of the respective V_{μ_X}. By Theorem 2(4), Φ_X identifies $E_1^-|_{V_{\mu_X}}$ with $\Phi_X(V_{\mu_X}) \cap E_1^-|_{V_\lambda}$. Hence

$$\Phi_A\big(B^{(\mu_A)}_{j_0(\mu_A)} V_{\mu_A}\big) \;\subseteq\; E_1^-|_{V_\lambda}, \qquad \Phi_B\big(B^{(\mu_B)}_{j_0(\mu_B)} V_{\mu_B}\big) \;\subseteq\; E_1^-|_{V_\lambda}\,.$$

Finally, each outer factor $(T_j{+}1)$ for $j \in \{j_0 - 1, j_0, \ldots, n - 2\}$ has index $j \geq j_0 - 1 = q + 2 \geq 5$ (since $q \geq 3$). Each such T_j commutes with T_1 (index gap $j - 1 \geq 4$), so each $(T_j{+}1)$ preserves $E_1^-|_{V_\lambda}$. Multiplying these together still preserves E_1^-. □ □

Corollary 15 (Rank-zero)**.** $\Pi^{S_n}|_{V_\lambda} = B_2^{(\lambda)} V_\lambda = 0$.

Proof. From Theorem 14, $B_{j_0}^{(\lambda)} V_\lambda \subseteq E_1^-$. Iterating downward: $B_{j_0-1}^{(\lambda)} V_\lambda = R'_{j_0-1} B_{j_0}^{(\lambda)} V_\lambda \subseteq R'_{j_0-1} E_1^- = 0$, since R'_{j_0-1} ends with $(T_1{+}1)$ on the right (and $j_0 - 1 \geq 2$). Successive iterations preserve 0, giving $B_2^{(\lambda)} V_\lambda = 0$. □ □

Theorem 16 (Upper bound)**.** *For $\lambda = (4, 3, 1^q)$ at every $q \geq 3$,*

$$\dim B_{j_0}^{(\lambda)} V_\lambda \ \leq \ 5.$$

The upper bound is structural for $q \geq 5$ and mixed structural-computational for $q \in \{3, 4\}$.

Proof. From the proof of Theorem 14,

$$B_{j_0}^{(\lambda)} V_\lambda \ = \ (T_{j_0-1}{+}1) \cdots (T_{n-2}{+}1) \Big[\Phi_A \big(B_{j_0(\mu_A)}^{(\mu_A)} V_{\mu_A} \big) + \Phi_B \big(B_{j_0(\mu_B)}^{(\mu_B)} V_{\mu_B} \big) \Big].$$

Each Φ_X is injective, so $\dim \Phi_X(B_{j_0(\mu_X)}^{(\mu_X)} V_{\mu_X}) = \dim B_{j_0(\mu_X)}^{(\mu_X)} V_{\mu_X}$. The sum of dimensions in the bracket is therefore at most

$$\dim B_{j_0(\mu_A)}^{(\mu_A)} V_{\mu_A} + \dim B_{j_0(\mu_B)}^{(\mu_B)} V_{\mu_B} \ = \ 2 + 3 \ = \ 5,$$

using Theorems 6(2) and 4(1). Multiplying by the outer factors $(T_j{+}1)$ cannot increase the dimension. Hence $\dim B_{j_0}^{(\lambda)} V_\lambda \leq 5$.

For $q \geq 5$, both summand dimensions are structural; for $q \in \{3, 4\}$, $\dim B^{(\mu_A)} = 2$ at $\mu_A = (3, 3, 1^{q-1})$ with $q - 1 \in \{2, 3\}$ is computational (Theorem 6(2)), and $\dim B^{(\mu_B)} = 3$ at $\mu_B = (4, 2, 1^{q-1})$ with $q - 1 \in \{2, 3\}$ (i.e. $|\mu_B| \in \{8, 9\}$) is also computational (Theorem 4(1)). □ □

6 Computational verification

Direct dim test for $\lambda = (4, 3, 1^q)$. Using `compute_dim_B.py` (May-13 multi-corner paper's script), mod $P = 100\,003$ at $q_{\text{spec}} = 11$:

λ	$\dim V_\lambda$	$\dim B_{j_0}^{(\lambda)} V_\lambda$
$(4, 3, 1, 1, 1)$ $(q = 3,\ n = 10)$	525	5
$(4, 3, 1, 1, 1, 1)$ $(q = 4,\ n = 11)$	1100	5

Both match the conjectural $f^{(3,2)} = 5$, agreeing with the structural upper bound and the May-13 multi-corner closed form.

Same-row part dimension. For the S-piece: $\dim S = f^{\nu_2} = f^{(4,1^{q+1})} = \binom{q+4}{3}$. At $q = 3$: $\dim S = 35$. At $q = 4$: $\dim S = 56$. Despite this large dimension, Lemma 13 forces $P_{j_0,n-1} S = 0$.

7 Open problems and follow-up

Structural lower bound: $\dim B \geq 5$. The upper bound $\dim B \leq 5$ is matched by the computational lower bound $\dim B \geq 5$ at $q \in \{3, 4\}$. A structural lower bound would construct 5 explicit vectors $F_{X,Y}$ in $B_{j_0}^{(\lambda)} V_\lambda$ (indexed by an SYT of μ_A or μ_B combined with a corner-pair label) and separate them via position-of-n / position-of-$(n-1)$ projections, analogous to the May-12-Eve3 lower-bound argument for $(4, 2, 1^*)$. We defer this to a follow-up paper.

Dim formula at $q \in \{3,4\}$, fully structural. Theorem 16's upper bound at $q \in \{3,4\}$ relies on the computational dim $= 2$ for $\mu_A = (3,3,1^{q-1})$ at $q-1 \in \{2,3\}$ *and* on the computational dim $= 3$ for $\mu_B = (4,2,1^{q-1})$ at $|\mu_B| \in \{8,9\}$. Structural proofs of either dim, at sub-shape sizes 8–9, would propagate this paper's upper bound to fully structural at the corresponding q values. The smaller sub-shapes are also closer to the rank-zero boundary, so a separate small-case treatment may be required.

Beyond $(4,3,1^*)$: the $(k,3,1^*)$ family. The proof of Lemma 13 (same-row-dies for λ) applies to any $(k,3,1^q)$ at $k \geq 4$ with $\lambda_2 = 3$ and $\lambda_3 = 1$: the sub-shape $\nu_2 = (k,1^{q+1})$ is a hook and Theorem 9 applies. The four-branch reduction (Theorem 11) and the μ_X identifications analogously give the upper bound $\dim B^{(\lambda)}_{j_0} V_\lambda \leq f^{(k-1,2)}$ *provided* the recursive sub-shape inputs ($\mu_A = (k-1,3,1^{q-1})$, $\mu_B = (k,2,1^{q-1})$, $\mu_C = (k-1,2,1^q)$) themselves have known dim/j-formulas. For $k = 5$: $\mu_A = (4,3,1^{q-1})$ is the family of this paper (at one-smaller q), giving a recursive bootstrap upward.

The general program. Combining the same-row-dies framework with the multi-corner recursion gives a roadmap for proving the closed form $\dim B^{(\lambda)}_{j_0} V_\lambda = f^{\lambda^{(\geq 2)}}$ structurally across the stable regime: each step adds a new shape class to the proven j-formulas, which in turn unlocks new sub-shape inputs via the same-row-dies framework. This paper closes $(4,3,1^*)$; the next natural targets are $(5,3,1^*)$ and $(5,4,1^*)$, which appear within reach using the same toolkit.

8 Honest status

What we proved. For $\lambda = (4,3,1^q)$ at every $q \geq 3$:

1. The four-branch reduction structurally identifies $B^{(\lambda)}_{j_0} V_\lambda$ with (outer factors) $\cdot\, \Phi_A(B^{(\mu_A)})$ + (outer factors) $\cdot\, \Phi_B(B^{(\mu_B)})$ (Theorem 11 + Lemmas 12, 13). This reduction itself is fully structural at $q \geq 3$: the two zero pieces (μ_C-leakage and same-row-dies) use only May-12-today, SRD-421-Theorem-14, and the Shift Lemma all-hooks j-formula.
2. The j-formula at sharp index: $B^{(\lambda)}_{j_0} V_\lambda \subseteq E_1^-|_{V_\lambda}$, hence rank-zero $\Pi^{S_n}|_{V_\lambda} = 0$ (Theorem 14, Corollary 15). Structural at $q \geq 5$; conditional on the computational μ_B-input at $q \in \{3,4\}$.
3. The upper bound $\dim B^{(\lambda)}_{j_0} V_\lambda \leq 5$ (Theorem 16), structural at $q \geq 5$ and conditional on computational dim inputs from μ_A, μ_B at $q \in \{3,4\}$.

What we did not prove. The matching structural lower bound $\dim \geq 5$. Computationally, $\dim = 5$ at $q \in \{3,4\}$ from `compute_dim_B.py`.

What this closes. This is the first $r = 2$ same-row family beyond $(4,2,1^*)$ where the structural j-formula at sharp index is established. It demonstrates that the May-13 same-row-dies framework, combined with the prior j-formulas on hooks and small non-hook shapes, propagates across an infinite-q family. The proof method is robust and points toward $(5,3,1^*)$, $(k,3,1^*)$, and recursive bootstrapping up the multi-corner ladder.